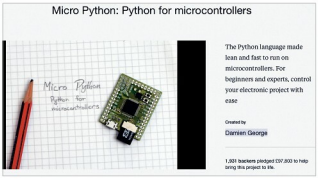

Figure1: MicroPython Kick-starter campaign

matures. The current board support list is given below.

- **pyboard**
  This is the official MicroPython board, which comes with full MicroPython support out-of-the-box. It features a STM32F405RGT6 ARM Cortex M4 based processor. MicroPython supports APIs for almost all hardware peripherals of the chip like GPIO, i2c, SPI, UART, etc.

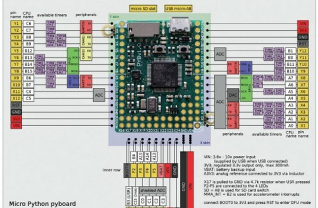

Figure 2: The PyBoard

- **ESP8266**
  MicroPython support for the low cost Wi-Fi development ESP8266 microcontroller is excellent. Wi-Fi and Internet accessibility are made available within MicroPython using a special network library. There's also a Web-based REPL that allows you to run MicroPython code on ESP8266 by using a Web browser within a local network.
- **BBC micro:bit**
  This is a great new board that is aimed at teaching kids computer programming. With MicroPython on the micro:bit, you can access the board's peripherals like LEDs, accelerometers, etc.

Note: The full list of supported boards can be found at the MicroPython website.

Figure 3: ESP826

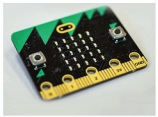

Figure 4: BBC micro:bit

Library stacks supported by MicroPython

Python supports a variety of libraries on desktop computers, but porting them directly to a microcontroller is not an easy task as they are not optimised for running on machines with less RAM. Instead of making existing useful Python libraries available in MicroPython, they are stripped in order to be better optimised to run on constrained hardware. The current libraries supported by MicroPython are named with the initial *u*, which stands for the Micro Library implementation:

- ucollections – collection and container types
- uhashlib – hashing algorithm
- uheapq – heap queue algorithm
- uio – input/output streams
- ujson – JSON encoding and decoding
- uos – basic 'operating system' services
- ure – regular expressions
- usocket – socket module
- ustruct – pack and unpack primitive data types
- utime – time related functions
- uzlib – zlib decompression

MicroPython-specific libraries

Functionality specific to the MicroPython implementation is available in the following libraries.

- machine — for functions related to the board
- micropython – to access and control MicroPython internals
- network — for network configuration
- uctypes – to access binary data in a structured way

Libraries specific to pyboard

The following libraries are specific to pyboard:

- pyb — for functions related to the board
- Time related functions
- Reset related functions
- Interrupt related functions

Note: The full list of supported libraries can be found at the MicroPython website.

Loading MicroPython on an ESP8266 board

ESP8266 is a cheap Wi-Fi development board. Since MicroPython exposes all the required APIs, including the networking stack, this chip is our preferred choice for learning how to load MicroPython. The ESP8266 raw chip